జాతీయ సాంకేతిక విజ్ఞాన సంస్థ ఆంధ్రప్రదేశ్ | राष्ट्रीय प्रौद्योगिकी संस्थान आंध्र प्रदेश
NATIONAL INSTITUTE OF TECHNOLOGY ANDHRA PRADESH
(An autonomous Institute under the aegis of Ministry of Education, Govt. of India)
सत्यमेव जयते
Ministry of Education
Government of India

SOLAR PANEL PERFORMANCE MONITORING SYSTEM

OUR TOPIC - SOLAR PANEL PERFORMANCE MONITORING SYSTEM

PV Analysis & Monitoring System – Explained:

The Solar Panel Performance Monitoring System using ESP32 and ADS1115 is designed to analyze the electrical and thermal performance of a 12V photovoltaic (PV) panel. The system measures voltage, current, and temperature using various sensors, with data processing and visualization handled through the ESP32 microcontroller.

PROCESSING PATHWAY

- The HCPL-7840 optical isolation amplifier from Broadcom is used to measure voltage safely in the system. It operates with 0.5% accuracy and has a 0–2V input range, making it suitable for power electronics and renewable energy applications.
- To step down the PV panel voltage, a voltage divider with 1MΩ and 100kΩ resistors is used. Additionally, an OPA333 operational amplifier from Texas Instruments stabilizes the low-voltage signals for accurate measurement.
- For current measurement, the ACS712 Hall Effect Current Sensor is used to measure DC current. It operates by detecting the magnetic field generated by the current flowing through the primary conductor, providing an isolated and accurate measurement.

LORA TRANSMISSION & RECEPTION

- The Ra-02 LoRa module, based on the SX1278 chip, is used for long-range, low-power wireless communication. It operates on 433MHz (default) and supports Chirp Spread Spectrum (CSS) modulation for reliable data transfer.

Transmission (ESP32 → LoRa Ra-02)

1. Data Collection: ESP32 reads voltage, current, and temperature data from sensors.
2. SPI Communication: The ESP32 sends data to the LoRa Ra-02 module using SPI.
3. LoRa Packet Formation: The data is structured into packets before transmission.
4. Frequency Setup: The LoRa module is configured to use a specific frequency (e.g., 433 MHz).
5. Transmission: The LoRa module sends the packet wirelessly to the receiver

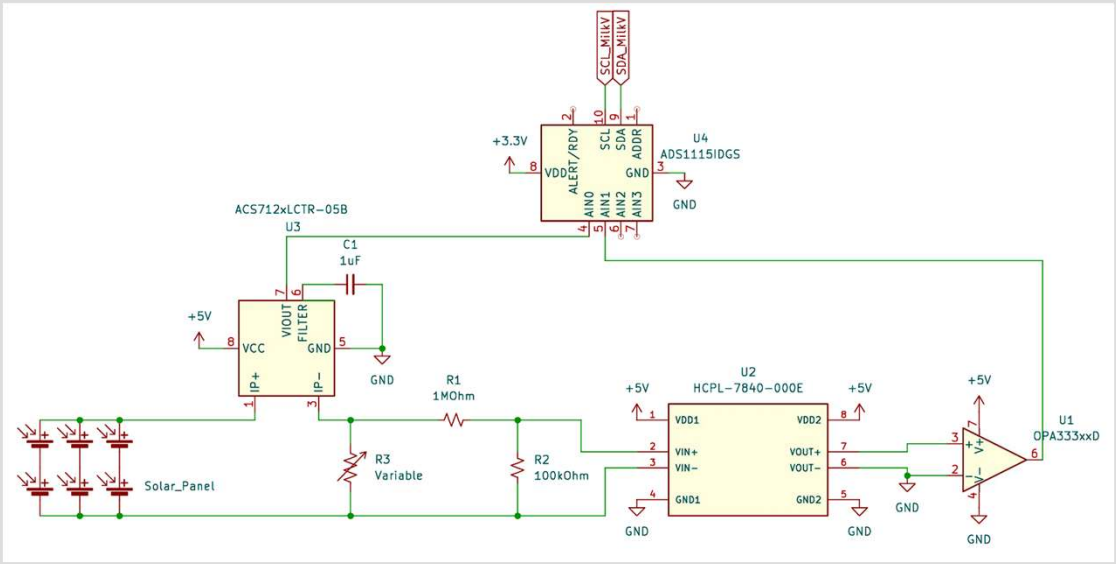
Reception (LoRa Ra-02 → PC via UART)

1. LoRa Signal Reception: The receiver module captures the transmitted LoRa signal.
2. Packet Decoding: The LoRa module decodes the received packet.
3. MCU Processing: A microcontroller (Arduino/ESP32) reads the data from Ra-02 via SPI.
4. Data Forwarding: The MCU forwards the received data to the PC via UART (FT232RL USB-to-Serial adapter).
5. PC Processing: The PC reads and processes the data using a serial terminal or Python script.

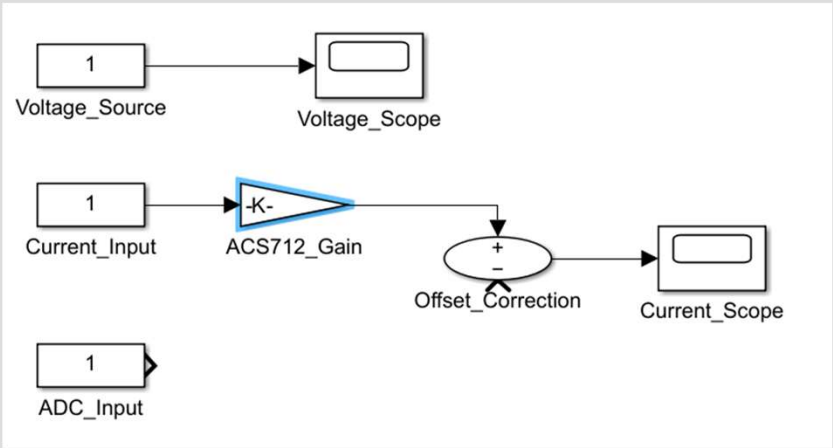
KEY COMPONENTS

- Microcontroller: ESP32
- Voltage & Current Measurement:
  - ADS1115 (16-bit ADC) for I2C-based voltage/current sensing
  - HCPL-7840, OPA333, and B0505S for voltage scaling (2V output)
  - CYCT03-L20 Hall Effect DC Current Sensor and ACS712 for current sensing
- Temperature Sensing: MAX31865 RTD-to-Digital Converter via SPI
- Ra-02 SX1278 LoRa Module (433MHz)
- ESP32 as an SPI to UART Interface
- Communication Interfaces: I2C and SPI configured on ESP32, UART from ESP32 to PC
- Software Development: Python for data acquisition
- Data Analysis & Visualization: MATLAB

CIRCUIT DIAGRAM



Physical Circuit Schema



Simulink .mat schema for script

SCRIPT SETUP

```
% MATLAB Analysis Script for PV Monitoring System
% This script processes voltage, current, and power data from Simulink

clear; clc; close all;
```

Load Simulink Data

```
load('PV_Monitoring_System.mat');


time = Voltage_Scope.time;
voltage = Voltage_Scope.signals.values;
current = Current_Scope.signals.values;
```

Noise Filtering (Moving Average Filter)

```
windowSize = 10;
voltage_filtered = movmean(voltage, windowSize);
current_filtered = movmean(current, windowSize);
```

Power Calculation

```
power = voltage_filtered .* current_filtered;
```

Plot Results

```
figure;

subplot(3,1,1);
plot(time, voltage_filtered, 'b', 'LineWidth', 1.5);
title('Filtered Voltage Over Time');
xlabel('Time (s)');
ylabel('Voltage (V)');
grid on;

subplot(3,1,2);
plot(time, current_filtered, 'r', 'LineWidth', 1.5);
title('Filtered Current Over Time');
xlabel('Time (s)');
ylabel('Current (A)');
grid on;

subplot(3,1,3);
plot(time, power, 'g', 'LineWidth', 1.5);
title('Power Output Over Time');
xlabel('Time (s)');
ylabel('Power (W)');
grid on;
```

Efficiency Calculation (Optional - if reference values exist)

Reference Power (e.g., Theoretical max power from PV panel)

```
P_ref = 50;
efficiency = (mean(power) / P_ref) * 100;

fprintf('Average Power Output: %.2f W\n', mean(power));
fprintf('System Efficiency: %.2f%%\n', efficiency);
```